III.4 The Simplex Method – Exceptions

Vocabulary: Changing from one simplex tableau to the next is called *pivoting,* or a *pivot.*

Remember in pivoting, we make one nonbasic variable x_i to be a basic variable (we say: "x_i enters the basis"), and in turn, we make one basic variable x_j to be nonbasic (we also say: "x_j leaves the basis")

A <u>feasible solution</u> is a solution

$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ of all the <u>constraints</u>

When applying the simplex algorithm, we sometimes may encounter problems: Explain by examples:

1. Unboundedness. Consider the following simple LP problem:

$$
\begin{aligned}
\text{Maximize} \quad & z = x_1 \\
\text{subject to} \quad & x_1 - x_2 \le 1 \\
& -x_1 + x_2 \le 2 \\
& x_1, x_2 \ge 0
\end{aligned}
\qquad (3.7)
$$

sketch the line $x_1 - x_2 = 1$

or $\boxed{x_2 = x_1 - 1}$

Slope $m = 1$

sketch the line $-x_1 + x_2 = 2$

or $\boxed{x_2 = x_1 + 2}$

Slope $m = 1$

$\Downarrow$

both lines have the same slopes $m = 1$

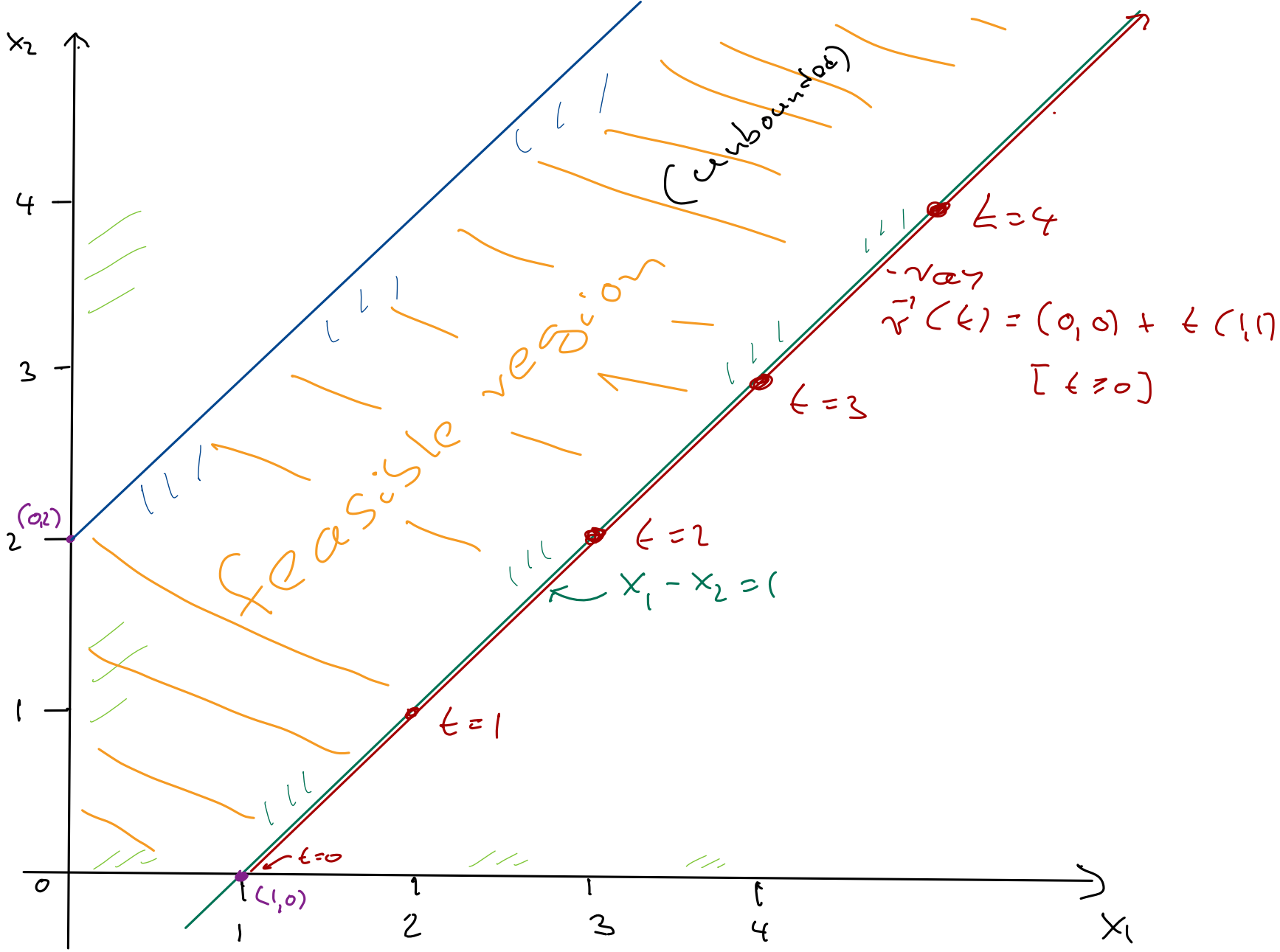
So are parallel

In canonical form, the constraints become equality constraints:

$$
(*) \quad \begin{cases}
x_1 - x_2 + x_3 = 1 \\
-x_1 + x_2 + x_4 = 2 \\
x_1, x_2, x_3, x_4 \ge 0
\end{cases}
$$

slack variables

The feasible region is unbounded:


x_2
x_1
0
1
2
3
4
(0,2)
(1,0)
feasible region
(unbounded)
$x_1 - x_2 = 1$
$t=0$
$t=1$
$t=2$
$t=3$
$t=4$
ray
$\vec{r}(t) = (0,0) + t(1,1)$
$[t \geq 0]$

Let's see what happens in the simplex algorithm when the feasible region is unbounded.

from (*)

feasible basic solution $(x_1, x_2, x_3, x_4) = (0, 0, 1, 2)$

The initial simplex tableau is:

basic variables

$$x_3 = 1 - x_1 + x_2 \qquad (3.7a)$$

$$x_4 = 2 + x_1 - x_2 \qquad (3.7b)$$

$$z = x_1 \qquad (3.7c)$$

can increase x_1

$\boxed{x_1 \le 1}$

$x_1 \ge 0$ arbitrary

The basis is $B = \{3, 4\}$, and the basic variables are x_3, x_4.

<u>First pivot:</u> Make x_1 a basic variable. In (3.7b), x_1 can be arbitrary. However, in (3.7a) we need $x_1 \le 1$.

Thus we use (3.7a) to make x_1 a basic variable ("x_1 enters the basis") and make x_3 a nonbasic variable ("x_3 leaves the basis")

(3.7a): gives $x_1 = 1 + x_2 - x_3$ (3.8a)

Also, (3.7b) becomes $x_4 = 2 + \underbrace{(1 + x_2 - x_3)}_{x_1} - x_2 = 3 - x_3$ (3.8b)

By (3.7c), $z = x_1 = 1 + x_2 - x_3$ (3.8c)

feasible basic solution

The next simplex tableau is

$(x_1, x_2, x_3, x_4) = (1, 0, 0, 3)$

$$x_1 = 1 + x_2 - x_3 \qquad (3.8a)$$ ⇐ $x_2 \geq 0$ arbitrary

$$x_4 = 3 \qquad - x_3 \qquad (3.8b)$$ ⇐ there is no x_2

$$z = 1 + x_2 - x_3 \qquad (3.8c)$$

↳ can increase x_2

<u>2nd pivot:</u> Next we should make x_2 a basic variable. Note: Neither of equations (3.8a) and (3.8b) restricts the size of x_2 ! So x_2 can be <u>any non-negative</u> number. We can't continue.

this indicates that no max of z exists: x_2 can be as large as we wish. $z = 1 + x_2 - x_3$ can be as large as we wish.

Let us set $x_2 = t, \quad (t \geq 0)$ and $x_3 = 0$, then also

$$x_1 = 1 + t \qquad \text{(by (3.8a))}$$

$$x_4 = 3 \qquad \text{(by (3.8b)),}$$

We have found infinitely many feasible solutions of the form

$$(**) \qquad (x_1, x_2, x_3, x_4) = (1,0,0,3) + t(1,1,0,0) \qquad (t \geq 0)$$

for which the cost $z = 1 + t \to \infty$ as $t \to \infty$.

There is no max. value for the cost variable z!

Graphically, the whole ray $\mathbf{r}(t) = (1,0) + t(1,1) \qquad (t \geq 0)$ lies in the feasible region.

The simplex method fails because there is no maximum on the feasible region.

In general, if

1. the LP problem is in {standard form, or} canonical form
2. the feasible region is unbounded

then the cost function may have no max.

2. Degeneracy. Consider the following simple LP problem:

Maximize $\boxed{z = x_2}$

subject to

$$-x_1 + x_2 \le 0$$
$$x_1 \le 2$$
$$x_1, x_2 \ge 0 \qquad (3.9)$$

sketch $-x_1 + x_2 = 0$

$\boxed{x_2 = x_1}$

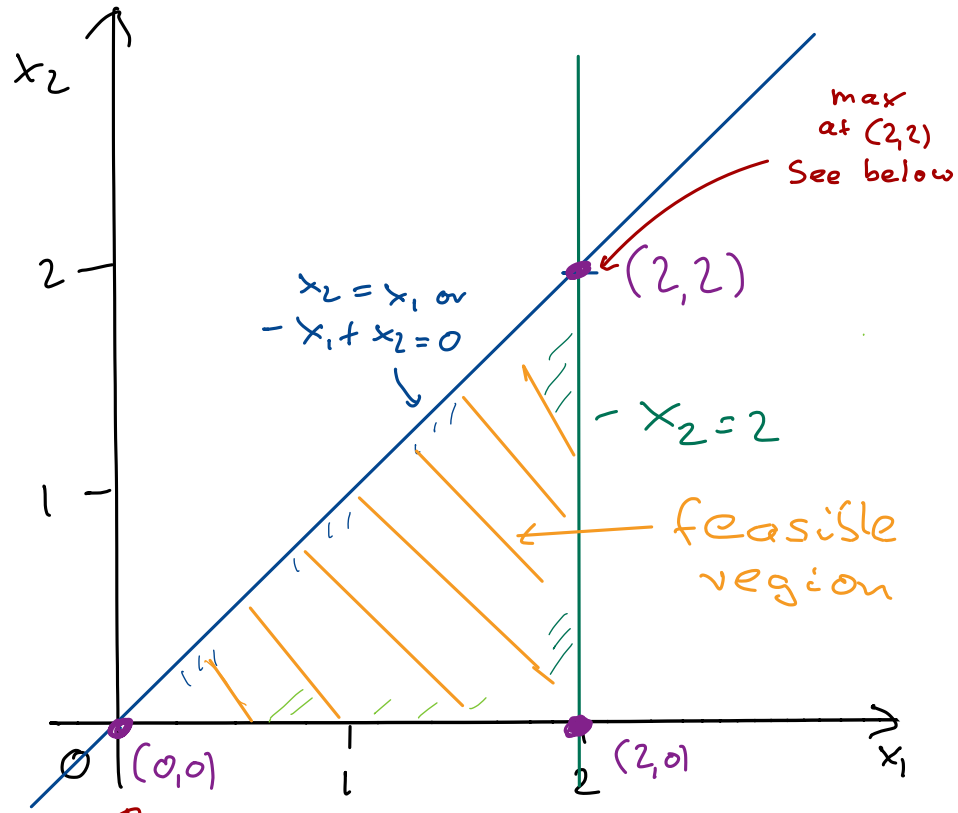
sketch $x_1 = 2$

In canonical (equational) form,

Constraints (*)

$$\begin{cases} -x_1 + x_2 + \boxed{x_3} = 0 \\ x_1 + \boxed{x_4} = 2 \\ x_1, x_2, x_3, x_4 \ge 0 \end{cases}$$

slack variables



we need to do a degenerate pivot before we can move from this point.

Construct the initial simplex tableau: from (*)

$$x_3 = \quad x_1 - x_2 \qquad (3.9a)$$

$$x_4 = 2 - x_1 \qquad (3.9b)$$

$$z = \quad x_2 \qquad (3.9c)$$

← x_2 cannot be increased because $x_1 = 0$

(z=0)

The basic solution is $(x_1, x_2, x_3, x_4) = (0,0,0,2)$ with basis $B = \{3, 4\}$.

By (3.9c), x_2 is the only candidate to could help increase the value of z. But (3.9a) shows that x_2 cannot be increased !

However, we can do a pivot (i.e. make x_2 a basic variable) without <u>decreasing</u> the value of z. This is called a *degenerate pivot*.

Solve (3.9a) for x_2 and get a new tableau:

"x_2 enters the basis"
"x_3 leaves the basis"

$$x_2 = \quad x_1 - x_3 \qquad (3.10a)$$

$$x_4 = 2 - x_1 \qquad (3.10b)$$

$$z = \quad x_1 - x_3 \qquad (3.10c)$$

← we can increase x_1 arbitrarily

← $x_1 \le 2$

(still z=0)
z is unchanged

change x_2 to $x_1 - x_3$

The basic solution is $(x_1, x_2, x_3, x_4) = (0,0,0,2)$, now with basis $B = \{2,4\}$.
(Note: This is the same basic solution as in (3.9), but with different basis B!)

We can now work with x_1. Equation (3.10b) restricts to $x_1 \le 2$, but (3.10a) does not restrict x_1. Use (3.10b) to switch x_1 and x_4: "x_1 enters the basis" "x_4 exits the basis"

$$x_1 = 2 - x_4 \qquad (3.11b)$$

Then (3.10a) becomes $x_2 = (2 - x_4) - x_3 = 2 - x_3 - x_4$ (3.11a)

We pivot to a new tableau:

$$z = x_1 - x_3$$
$$= (2 - x_4) - x_3$$
$$= 2 - x_3 - x_4$$

$$x_1 = 2 \qquad - x_4 \qquad (3.11b)$$

$$x_2 = 2 - x_3 - x_4 \qquad (3.11a)$$

$$z = 2 - x_3 - x_4 \qquad (3.11c)$$

($z = 2$, is increased)

which gives the optimal basic solution $B = \{1,2\}$, x_1, x_2 are basic

$$\mathbf{x}^T = (x_1, x_2, x_3, x_4) = (2,2,0,0),$$

$\boxed{\text{max of } z \text{ is } z = 2}$

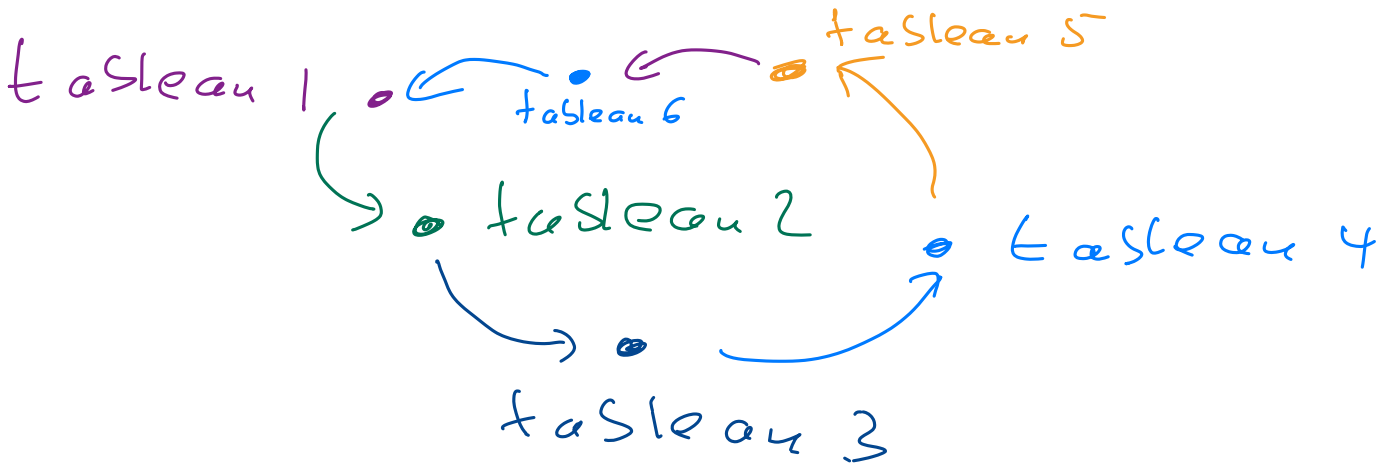
(Because all slack variables are zero, we are sure to have found an optimal solution.)

A *degenerate pivot* is a pivot that does not increase the value of the cost variable z, but keeps it the same.

As seen above, sometimes one has to perform one (or several) degenerate pivots.

It can happen (very rarely) that after many degenerate pivots, one returns to a previously tableau and is stuck; this is called cycling.

(we don't worry about this. It happens rarely)



Cycling: After many degenerate pivots, we return to the original tableau.

[Can only happen when z does not increase, i.e. degenerate pivot]

Example: Solve the following LP in standard form:

Maximize $z = 2x_1 + 2x_2 + x_3$

subject to

$$\begin{aligned} 2x_1 \quad\quad + x_3 &\le 4 \\ x_1 + x_2 \quad\quad &\le 1 \\ x_1 \quad\quad + x_3 &\le 1 \\ x_1, x_2, x_3 &\ge 0 \end{aligned}$$

Sol: In canonical form, the constraints are:

$$\begin{cases} 2x_1 + x_3 + \boxed{x_4} = 4 \\ x_1 + x_2 + \boxed{x_5} = 1 \\ x_1 + x_3 + \boxed{x_6} = 1 \\ x_1, x_2, x_3, x_4, x_5, x_6 \ge 0 \end{cases}$$

slack variables = basic variables for the initial tableau

The initial tableau is

$$
\begin{array}{lrcl}
(a) & x_4 & = & 4 - 2x_1 - x_3 \\
(b) & x_5 & = & 1 - x_1 - x_2 \\
(c) & x_6 & = & 1 - x_1 - x_3 \\
\hline
(d) & z & = & 2x_1 + 2x_2 + x_3
\end{array}
$$

$\leftarrow x_1 \leq 4$

$\leftarrow x_1 \leq 1 \Rightarrow$ use this equation

$\leftarrow x_1 \leq 1$

↳ make x_1 a basic variable

"x_1 enters, x_5 exits"

<u>1st. pivot:</u>

(b) becomes: $\boxed{x_1 = 1 - x_2 - x_5}$ (1a)

Then (a) becomes:

$$x_4 = 4 - 2\left[1 - x_2 - x_5\right] - x_3 = 2 + 2x_2 - x_3 + 2x_5 \quad (1b)$$

and (c) becomes

$$x_6 = 1 - [1 - x_2 - x_5] - x_3 = x_2 - x_3 + x_5 \qquad (1c)$$

The cost function becomes

$$z = 2[1 - x_2 - x_5] + 2x_2 + x_3 = 2 + x_3 - 2x_5 \qquad (1d)$$

we obtain the tableau

$$\begin{array}{lll}
x_1 = 1 - x_2 - x_5 & (1a) & \\
x_4 = 2 + 2x_2 - x_3 + 2x_5 & (1b) & x_3 \le 2 \\
x_6 = x_2 - x_3 + x_5 & (1c) & x_3 \le 0 \\
\hline
z = 2 + x_3 - 2x_5 & (1d) &
\end{array}$$

↑ increment not possible by (1c)

<u>2nd pivot</u>

In the next pivot we want to increment x_3.

But (1c) shows that x_3 in the tableau can <u>not</u> be increased !!

So we do a degenerate pivot using (1c)

"x_3 enters the base, x_6 exits the base"

By (1c): $x_3 = x_2 + x_5 - x_6$ (2c)

(1a) remains unchanged

(1b): $x_4 = 2 + 2x_2 - [x_2 + x_5 - x_6] + 2x_5 = 2 + x_2 - x_5 + x_6$ (2b)

(1d) $z = 2 + [x_2 + x_5 - x_6] - 2x_5 = 2 + x_2 - x_5 - x_6$ (2d)

We get the new tableau:

$$x_1 = 1 - x_2 - x_5 \quad (2a)$$

$$x_3 = + x_2 + x_5 - x_6 \quad (2c)$$

$\leftarrow x_2$ unconstrained

$$x_4 = 2 + x_2 - x_5 + x_6 \quad (2b)$$

$\leftarrow x_2$ unconstrained

$$z = 2 + x_2 - x_5 - x_6 \quad (2d)$$

↳ we can increase x_2.

3rd pivot: By (2a), "x_2 enters, x_1 leaves"

(2a): $\boxed{x_2 = 1 - x_1 - x_5}$ (3a)

(2c): $x_3 = [1 - x_1 - x_5] + x_5 - x_6 = 1 - x_1 - x_6$ (3c)

(2b): $x_4 = 2 + [1 - x_1 - x_5] - x_5 - x_6 = 3 - x_1 - 2x_5 - x_6$ (3b)

(2d): $z = 2 + [1 - x_1 - x_5] - x_5 - x_6$

$= 3 - x_1 - 2x_5 - x_6$

Can't increas any variable. → Stop

Basic feasible solution: $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 1, 1, 3, 0, 0)$

z is largest, when $\boxed{x_1 = 0,\ x_2 = 1,\ x_3 = 1}$

and then $\boxed{z = 3}$

(Optimal solution)

3. Infeasibility. Consider an LP problem in canonical form:

Maximize $z = x_1 + 2x_2$

equality conditions already !!

subject to

$$x_1 + 3x_2 + x_3 = 4 \qquad (1)$$
$$2x_2 + x_3 = 2 \qquad (2)$$
$$x_1, x_2, x_3 \geq 0 \qquad (3)$$

See also the remark below

We have difficulties finding an initial feasible solution (i.e. (x_1, x_2, x_3) satisfying the constraints) required for the initial tableau. Because there are 2 equations, but 3 variables, we need a feasible solution (x_1, x_2, x_3) where ONE $x_i = 0$, which will be a nonbasic variable. The other two x_j will be the basic variables. [The following idea works when # of equation is < # of variables]

Idea: introduce artificial slack variables x_4, x_5 into the constraints, and change (1)-(3) to

$$x_1 + 3x_2 + x_3 + \boxed{x_4} = 4 \qquad (a)$$
$$2x_2 + x_3 \boxed{+ x_5} = 2 \qquad (b)$$
$$x_1, x_2, x_3\ x_4, x_5 \geq 0 \qquad (c)$$

Note: a feasible solution of (a)-(c) with $x_4 = 0, x_5 = 0$

will give a feasible solution of (1)-(3) !!!

We must find a feasible solution $(x_1, x_2, x_3, x_4, x_5)$ of these constraints, where $x_4 = 0$ and $x_5 = 0$ (because then (x_1, x_2, x_3) will be a feasible solution of (1)—(3). For this, we choose the cost function

$$z = -x_4 - x_5$$

we know that $z \leq 0$ always because $x_4, x_5 \geq 0$. So if Max $z = 0$ we can conclude that $x_4 = x_5 = 0$

(by (3a), $-x_4 - x_5 \leq 0$. So if we can solve the LP-problem

slack variables

Maximize	$z = -x_4 - x_5$	(0a)
subject to	$x_1 + 3x_2 + x_3 + \boxed{x_4} = 4$	(1a)
	$2x_2 + x_3 \boxed{+ x_5} = 2$	(2a)
	$x_1, x_2, x_3\ x_4, x_5 \geq 0$	(3a)

and the max of z is zero, i.. $-x_4 - x_5 = 0$, then it will follow that $x_4 = 0$ and $x_5 = 0$.

Step 1: solve the LP problem (0a)—(3a) .

The initial simplex tableau is:

From (1a): $x_4 = 4 - x_1 - 3x_2 - x_3$ $(x_1 \le 4)$ (2b)

From (1b) $x_5 = 2 \quad - 2x_2 - x_3$ (x_1 any) (3b)

Basic variables

$z = -x_4 - x_5$

express these in terms of the nonbasic variables x_1, x_2, x_3 (use (2a), (3a))

$z = -[4 - x_1 - 3x_2 - x_3] - [2 - 2x_2 - x_3]$

$$\boxed{z = -6 + x_1 + 5x_2 + 2x_3} \quad (4b)$$

1st pivot:

make x_1 enter the basis, and x_4 exit [use (2b)]

→ From (2b): $\boxed{x_1 = 4 - 3x_2 - x_3 - x_4}$ (2c)

→ (3b) is unchanged [there is no x_1]

→ From (4b): $z = -6 + [4 - 3x_2 - x_3 - x_4] + 5x_2 + 2x_3$

$z = -2 + 2x_2 + x_3 - x_4$ (4c)

New tableau:

$$x_1 = 4 - 3x_2 - x_3 - x_4 \qquad x_3 \le 4 \qquad (2c)$$

$$x_5 = 2 - 2x_2 - x_3 \qquad x_3 \le 2 \qquad (3b)$$

$$z = -2 + 2x_2 + x_3 - x_4 \qquad (4c)$$

2nd pivot:

Make x_3 enter basis.

$x_3 \le 2 \Rightarrow$ use (3b), x_5 will exit.

$$\boxed{x_3 = 2 - 2x_2 - x_5} \qquad (2d)$$

By (2c): $x_1 = 4 - 3x_2 - [2 - 2x_2 - x_5] - x_4$

$$\Rightarrow x_1 = 2 - x_2 - x_4 + x_5 \qquad (3d)$$

By (4c): $z = -2 + 2x_2 + [2 - 2x_2 - x_5] - x_4$

$$\Rightarrow z = -x_4 - x_5 \qquad (4d)$$

Tableau:

basic variables

$$x_1 = 2 - x_2 - x_4 + x_5 \qquad (3d)$$
$$x_3 = 2 - 2x_2 \qquad - x_5 \qquad (2d)$$
$$z = - x_4 - x_5$$

The optimal feasible basic solution of (0a) – (3a) is

$x_4 = 0,\ x_5 = 0$

$$(x_1, x_2, x_3, x_4, x_5) = (2, 0, 2, 0, 0) \quad \text{with } \underline{\underline{z = 0\ !!}}$$

<u>Step 2:</u> Solve the original system (1) – (3):

Because $x_4 = 0$, $x_5 = 0$, removing these from the above tableau, we get:

$$x_1 = 2 - x_2 \qquad (x_2 \le 2) \qquad (2a)$$
$$x_3 = 2 - 2x_2 \qquad (x_2 \le 1) \qquad (3a)$$
$$z = x_1 + 2x_2 \qquad \Leftarrow \text{original cost function} \qquad (4-)$$

Pivot:

$2x_2 = 2 - x_3$

$x_2 = 1 - \frac{1}{2}x_3$

↑ basic variable, eliminate, using (2a)

$z = [2 - x_2] + 2x_2$

$z = 2 + x_2$ (4e)

↑ we can increase x_2 using (3e)

In (3e), x_2 enters, x_3 leaves the basis

$$\boxed{x_2 = 1 - \frac{1}{2}x_3} \quad (2f)$$

(2e) becomes: $x_1 = 2 - \left[1 - \frac{1}{2}x_3\right] = 1 + \frac{1}{2}x_3$ (3f)

(4e) becomes: $z = 2 + 2\left[1 - \frac{1}{2}x_3\right] = 3 - x_3$ (4f)

The tableau becomes:

$x_1 = 1 + \frac{1}{2}x_3$ (3f)

$x_2 = 1 - \frac{1}{2}x_3$ (2f)

$z = 3 - x_3$ stop

Answer:
(to problem (1)-(5))

z is max, $z = 3$, when $x_3 = 0$, $x_1 = 1$, $x_2 = 1$

Remark: There is an alternative of solving (1) – (3) <u>without</u> introducing slack variables:

Constraints:

$$\begin{cases} x_1 + 3x_2 + x_3 = 4 & (1) \\ 2x_2 + x_3 = 2 & (2) \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

We eliminate x_3 from (1):

(1) – (2):

$$\boxed{x_1} + x_2 = 2 \qquad (1')$$

only present in (1')

$$2x_2 + x_3 = 2 \qquad (2)$$

only present in (2)

$$z = x_1 + 2x_2$$

Simplex tableau:

$$x_1 = 2 - x_2$$
$$x_3 = 2 - 2x_2$$
$$z = x_1 + 2x_2$$

– This is (2a), (3a), (4–) above, and solved the same way